

Tirth Shah

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EDUCATION

Master of Science in Computer Science (GPA: 4.00/4.00)

August 2024 – May 2026

California State University Los Angeles

Los Angeles, CA

Relevant Coursework: Advanced Data Structures & Algorithms, Big Data, Advanced Machine Learning & Deep Learning, Advanced Software Engineering, Advanced Web Programming

TECHNICAL SKILLS

Programming: Python, R, Java, GoLang, C, JavaScript, TypeScript, PHP, SQL, HTML/CSS

AI & Machine Learning: LangChain, LangSmith, GraphRAG, RAG, Agentic Workflows, PyTorch, TensorFlow, BERT, LSTM, XGBoost, Random Forest, SVM, Genetic Algorithms

Data & Cloud: AWS (Bedrock, SageMaker), Azure, GCP, Snowflake, BigQuery, Hadoop, Neo4j, PostgreSQL, MongoDB, Tableau, PowerBI, scikit learn

Web & Frameworks: React, Node.js, REST APIs, WebSockets, JWT, React Native, Firebase

Tools & DevOps: Docker, Git, CI/CD, Traefik, Nginx, Cloudflare, n8n, Langflow, Groq, Astra DB, GridSearchCV

EXPERIENCE

AI Developer - ONLC Training Centers

June 2025 – Present

- Engineered an end to end eLearning platform suite in Python and TypeScript, accelerating content production by 70% across 5 teams via integrated Heygen and ElevenLabs APIs.
- Spearheaded DevOps infrastructure with CI/CD pipelines, Traefik and Nginx load balancers, and a Cloudflare security layer, standardizing deployments across production services.
- Architected a production RAG system on LangChain, GraphRAG, and Neo4j serving over 1,000 queries per day at under 2s latency, driving AI adoption for over 1,000 learners.

Research Assistant - California State University Los Angeles

September 2024 – May 2026

- Pioneered SAGAFS GA (Species Adaptive Genetic Algorithm for Feature Selection), reaching a 94% F1 score by optimizing over 20 ML models across 18 biomedical datasets.
- Accelerated drug discovery screening by 75% through SSOCF crossover and dynamic mutation rates, analyzing over 10,000 sequences in under 5 minutes.
- Authored research for an IEEE international conference demonstrating a 60% computational reduction, and open sourced the framework, which was adopted by an external research lab.

Data Science Intern - Prayosha Food Services Pvt. Ltd.

January 2024 – May 2024

- Improved demand forecasting accuracy by 23% using an XGBoost ensemble with Bayesian optimization over historical transaction data.
- Deployed 3 deep learning models (LSTM, CNN, Transformer) achieving 91% accuracy for inventory optimization, backed by distributed Hadoop processing.

ACADEMIC PROJECTS AND ACTIVITIES

Project Portfolio: [byTirth.vercel](#)

Agentic AI Application Development (Langflow, Groq, Astra DB, React, Node)

January 2025 – February 2025

- Built a multi agent RAG system reaching 95% retrieval accuracy by orchestrating a Langflow pipeline with Groq LLM inference, Astra DB vector storage, and Nvidia embeddings for PDF summarization.
- Cut development time by 40% with drag and drop workflows for automated web search and database querying, delivered through a full stack React and Node interface with WebSocket streaming and JWT auth.

NeuzNow: Cross Platform News App (React Native, Firebase, REST APIs)

January 2022 – May 2022

- Developed a full stack React Native application with a responsive UI for iOS and Android, delivering real time news aggregation with categorized content browsing.
- Integrated Firebase Cloud Messaging for push notifications and architected a modular component structure with a RESTful API layer for seamless news data synchronization.

Extracurricular Activities & Achievements

- Served as an active member of the CSULA AI/ML Club organizing workshops and tech talks, and as technical mentor at a university hackathon guiding teams on ML implementation.
- Volunteered over 100 hours teaching Python and data science fundamentals to underrepresented high school students through community outreach programs.